SpaceX Starlink Gen 3 Satellite Rigging Specification

Pennine 90mph Gale Force Wind Loading, Non-Drill Chimney Lashing & BS 6701 Cable Integration

1. Executive Engineering Summary

This technical specification defines the physical installation methodology for SpaceX Starlink Gen 3 and Gen 2 terminals across exposed UK upland residential structures. Authored by Gary Pearce, Lead Infrastructure Engineer at ADI Communications (Direct Mobile: 07860 244 274, 25+ years hands-on field craftsmanship), this document details structural wind-load mitigation, non-drill stainless chimney lashing cradles, and weather-sealed cavity wall drops complying with UK BS 6701 standards.

2. Structural Wind-Load & Shear Calculations (90mph Gale Rating)

Under BS EN 1991-1-4 (Eurocode 1: Actions on structures — General actions — Wind actions), a rectangular Starlink Gen 3 flat panel (0.22 m² surface area) experiences significant dynamic drag forces in Northern UK Pennine micro-climates:

Wind Speed (mph)	Velocity (m/s)	Dynamic Pressure (N/m²)	Drag Force (N)	Structural Mounting Requirement
45 mph (Gale)	20.1 m/s	247 N/m²	65 N	Standard 1.5-inch alloy standoff
70 mph (Severe Gale)	31.3 m/s	599 N/m²	158 N	Heavy-duty galvanised Pennine storm bracket
90 mph (Storm Force 10)	40.2 m/s	990 N/m²	261 N	Twin stainless steel chimney lashing kit with corner mitres
105 mph (Extreme Gust)	46.9 m/s	1,349 N/m²	356 N	10-gauge structural steel brackets with M10 masonry sleeves

3. Non-Drill Chimney Lashing Protocol

- **Zero Masonry Penetration:** Never drill into Victorian lime mortar or single-skin chimney brickwork, which causes frost heave and structural spalling.
- **Dual Steel Wire Cradles:** High-tensile 7-strand galvanised wire wrapped around the chimney perimeter, secured with heavy-duty J-bolts and tension ratchets.
- **Corner Mitre Angle Protectors:** Formed steel corner plates distribute mechanical clamping pressure evenly across 4 masonry corners, preventing edge-cracking.
- **External Solid Copper Cat6:** UV-stabilised high-grade shielded cable with a 150mm downward drip loop before masonry ingress.

Rule 45 Reference & Verification: *Full interactive wind-load calculators, obstacle cone diagnostic tools, and complete case studies are available on the ADI Communications Technical Portal:* ***https://adicommunications.co.uk/news/starlink-chimney-mount-vs-roof-apex/*** *and the Google Shield Companion Hub at* ***https://tvandhomesecurity.blogspot.com/****.*